NATIONAL JUNIOR COLLEGE
SH2 PRELIMINARY EXAMINATION
Higher 3

CANDIDATE NAME

SUBJECT CLASS

REGISTRATION NUMBER

PHARMACEUTICAL CHEMISTRY 9812/01

Paper 1 **Fri 18 Sep 2015**

Additional Materials: Answer Paper **2 hours 30 minutes**

Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, subject class and registration number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer any **five** questions.
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.
You may use a calculator.
You are reminded of the need for clear presentation in your answers.

This paper consists of **17** printed pages.

1 Nature is full of wonders. It is an excellent source of new drugs, or commonly, of precursors to drugs. Of the top twenty bestselling drugs in 1999, nine of them were derived from natural products. Almost 40 % of the 520 new drugs approved for the drug market between 1983 and 1994 were natural products or derived from natural products. Greater than 60 % of the anti-cancer and anti-infective agents that are on the market or in clinical trials are of natural products origins or derived from natural products. This may be the result of the inherent nature of these chemicals to act in defense of their producing organisms; for instance, a fungal natural product might be produced against bacteria or other fungi or against the cell replication of foreign organisms.

An extract of European mistletoe, *viscum album*, contains a series of pharmacologically active proteins known as viscotoxins. They are currently being studied as potential drugs in cancer treatment, which includes leukaemia and Yoshida sarcoma.

The primary structure of one of them, **viscotoxin A_2**, was determined using oxidation to break disulfide linkages, followed by enzymatic digestion of the peptide with trypsin and chymotrypsin. The structures of the peptide fragments that were isolated from the two enzymatic cleavages are given below.

Peptide fragments from trypsin digestion:

Asn-Ile-Tyr-Asn-Thr-Cys-Arg (7 amino acids)
Phe-Gly-Gly-Gly-Ser-Arg (6 amino acids)
Ile-Ile-Ser-Ala-Ser-Thr-Cys-Pro-Ser-Tyr-Pro-Asp-Lys (13 amino acids)
Lys-Ser-Cys-Trp-Pro-Asn-Thr-Gln-His-Arg (10 amino acids)

Peptide fragments from chymotrypsin digestion:

Gly-Gly-Gly-Ser-Arg-Glu-Val-Cys-Ala-Ser-Leu (11 amino acids)
Ser-Gly-Cys-Lys-Ile-Ile-Ser-Ala-Ser-Thr-Cys-Pro-Ser-Tyr-Pro-Asp-Lys (17 amino acids)
Lys-Ser-Cys-Trp-Pro-Asn-Thr-Gln-His-Arg-Asn-Ile-Tyr (13 amino acids)
Asn-Thr-Cys-Arg-Phe (5 amino acids)

Viscotoxin A_2 contains 46 amino acid residues. It shares the same N- and C-terminus and has a molecular mass of 5023 $g mol^{-1}$.

(a) Find the primary structure of **viscotoxin A_2**. Briefly explain your reasoning. [2]

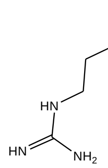
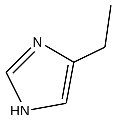
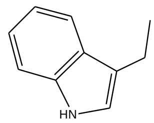
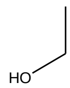
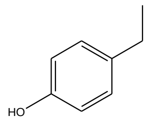
(b) Chemical hydrolysis usually suffices for the determination of primary structures of most proteins. If a protein strand is heated in concentrated hydrochloric acid at 37 °C for three hours, it is degraded into smaller protein fragments, which still allows us to piece back the original primary structure. However, chemical hydrolysis is not appropriate for analyzing the primary structure of **viscotoxin A_2**.

Explain why enzymatic hydrolysis is more suitable for accurate analysis of the primary structure of **viscotoxin A_2**.

[1]

(c) Electrophoresis is the routine method employed to separate the amino acid residues.

A sample of **viscotoxin $\mathbf{A_2}$** was completely hydrolyzed in 20 % hydrochloric acid for two days and then carefully neutralized. **Arginine**, **histidine**, **tyrosine**, **tryptophan** and **serine** were isolated and subjected to electrophoresis. These amino acids have the following isoelectric point (pI) values.

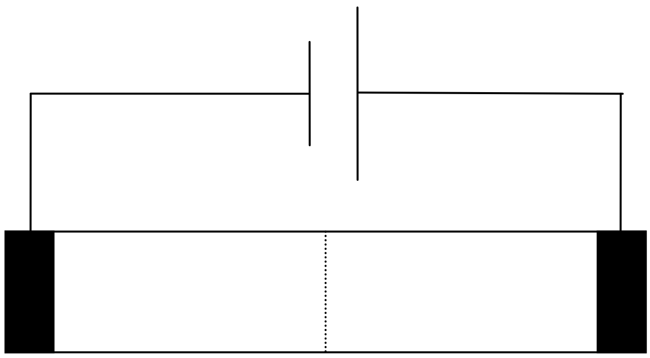
Amino acid (Abbreviation)	**Arginine (Arg)**	**Histidine (His)**	**Tryptophan (Trp)**	**Serine (Ser)**	**Tyrosine (Tyr)**
pI value	10.76	7.59	5.89	5.68	5.66
Structure of R group					

(i) Rationalize the relative pI values between histidine, arginine and tryptophan.

(ii) Copy the setup below on your writing paper and indicate the relative positions of the five amino acids when the electrophoresis buffer is set at

(I) pH = 5.89
(II) pH = 7.59

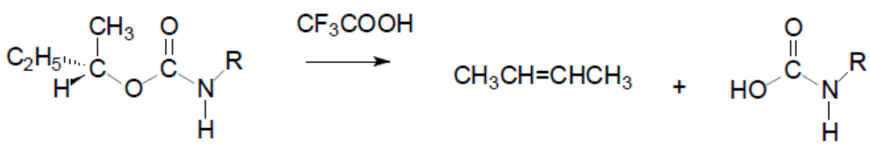
Use the abbreviations to represent the amino acids. For example, ‘tyrosine’ is to be represented by ‘Tyr’.



starting point

[7]

(d) The reaction scheme below illustrates a bimolecular elimination reaction, one of the many methods to synthesize amino acids.

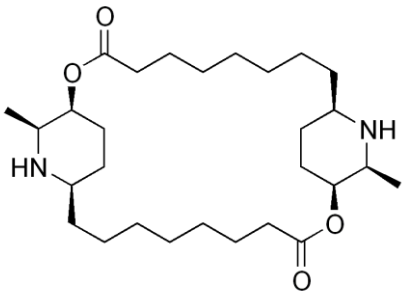


compound **X**

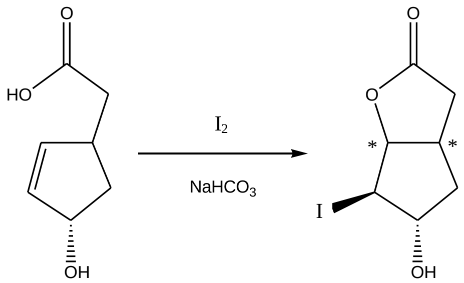
Using Newman projection, draw the mechanism of the above reaction and deduce the more favorable geometric isomer of the alkene formed.

[3]

(e) Carpaine is one of the major alkaloid components of papaya leaves which is studied for its cardiovascular effects.



A novel lactone synthesis involves the use of iodine in a process termed "iodolactonization". It consists of the electrophilic addition of iodine to the alkene, followed by the formation of the lactone.



(i) Outline a mechanism for the iodolactonization reaction. In your mechanism, you should indicate the stereochemistry of the (*)-labeled chiral carbons where relevant.

Using the iodolactonization reaction, carpaine can be synthesized according to the synthetic route below:

$$2\ \mathbf{K} \xrightarrow[-2\ HI]{1.\ I_2\quad 2.\ NaHCO_3} \mathbf{L} \xrightarrow[-2\ HI]{} \text{carpaine}$$

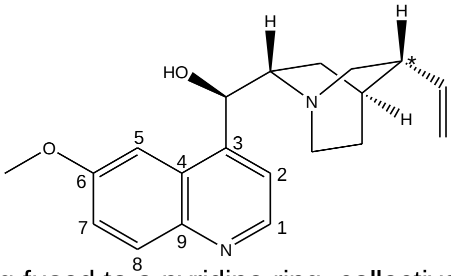
(ii) Draw the structures of **K** and **L**, showing the correct stereochemistry.

[7]

[Total: 20]

2 Quinine is the first anti-malarial drug discovered, first appearing in the seventeenth century. Although it is no longer recommended as a first-line medicine for malaria, it is still popularly found in tonic water or in the popular gin and tonic cocktail.

The structure of quinine is shown below:



Quinine has a benzene ring fused to a pyridine ring, collectively called a quinoline ring, as well as a bicyclic tertiary amine, collectively called a quinuclidine ring.

(a) **(i)** Using the R/S notation, identify the stereochemistry of the C atom labelled *.

(ii) Explain why nitration is unlikely to occur at C1, C2 and C8.

(iii) Explain why nitration occurs predominantly at C5 instead of C7, using relevant resonance structures to support your answer.

[6]

(b) Quinine was converted into a tosylate where the –OH group was replaced with a –O-Tos group (tosylate group) which is a better leaving group than –OH. The stereochemistry of the molecule was not affected. The tosylate was then heated strongly with sodium ethoxide in ethanol to obtain an alkene.

(i) Predict whether the E or the Z-isomer is formed. Explain your answer.

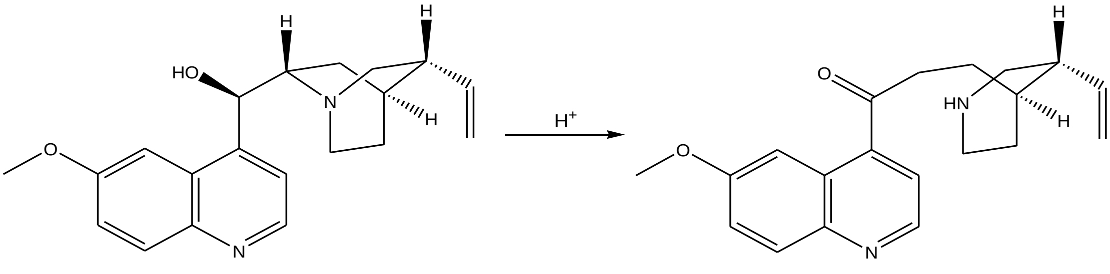
(ii) Explain the effect on the stereochemical purity of the product by changing the ethanol solvent to an aqueous solvent.

[4]

(c) The quinuclidine ring has a pK_b of 3.00 while trimethylamine has a pK_b of 4.13.

(i) Account for the greater basicity of quinuclidine.

(ii) In 1853, Louis Pasteur discovered that quinine would isomerize upon standing in aqueous acid.

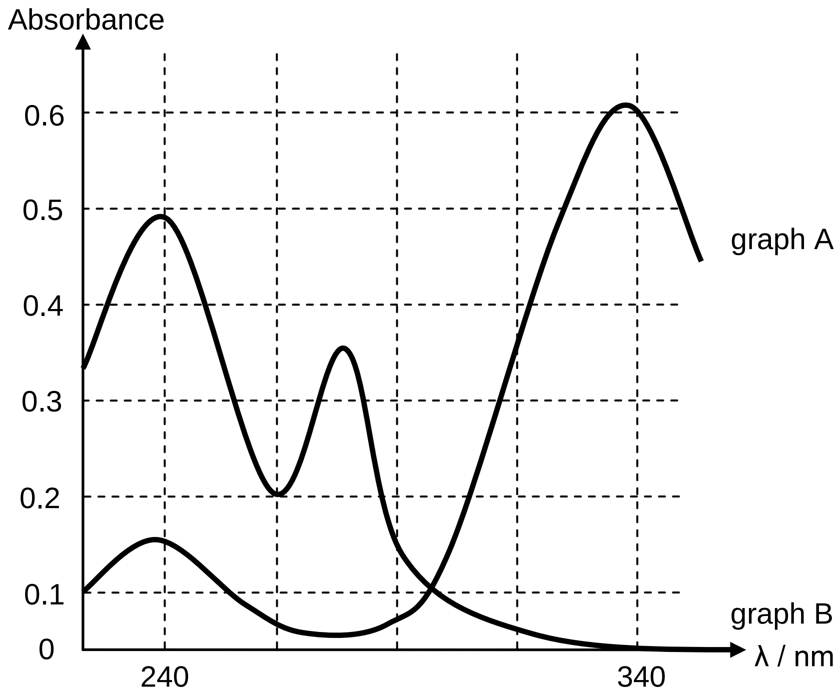


Outline a mechanism for this isomerization.

[4]

(d) Schweppes tonic water contains the following ingredients: carbonated water, citric acid, sodium benzoate, sugar syrup and quinine. It is usually sold in sizes of 175 m*l*.

A standard solution of 7.5×10^{-5} mol dm^{-3} of sodium benzoate and quinine produced the following UV spectrum.



A bottle of Schweppes tonic water was analyzed using with UV/VIS spectroscopy at 240nm and 340nm with 1.0 cm cuvettes. The absorbance recorded at each wavelength of light was:

Wavelength / nm	**Absorbance**
240	0.920
340	1.404

(i) Explain which compound would give rise to graph A and B

(ii) Determine the concentration of quinine in the drink.

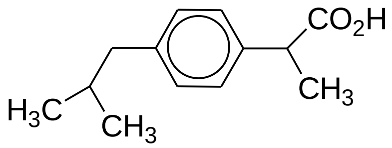
(iii) The recommended dosage for malaria treatment is two 300mg quinine tablets per dose. Calculate the number of Schweppes tonic water cans required to match this dosage. [M_r (quinine) = 324]

(iv) The molar extinction coefficient of quinine at 240nm is 4700 mol^{-1}dm^{3}cm^{-1}. Using your answer to (d)(ii), calculate the absorbance of sodium benzoate at 240nm and hence determine the concentration of sodium benzoate in the drink.

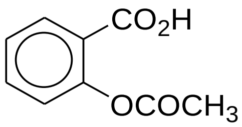
[5]

[Total: 20]

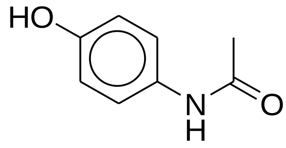
3 The following show the structural formula of some antipyretic pain killers.



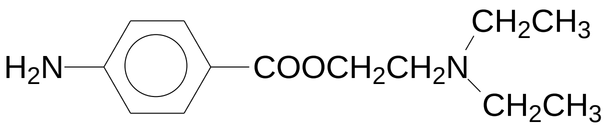
Ibuprofen



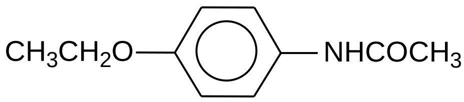
Aspirin



Paracetamol



Procaine



Phenacetin

When these compounds are ingested into the body, they are metabolized by enzymes in the liver.

(a) **(i)** Give two differences in the effects of aspirin and paracetamol on the body.

(ii) The primary metabolic step involves a **different mechanism** for each of the compounds listed below.

aspirin	phenacetin
ibuprofen	procaine

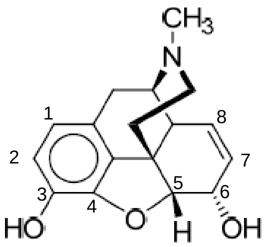
The four different mechanisms are listed below.

O-dealkylation	hydrolysis
N-dealkylation	aliphatic hydroxylation

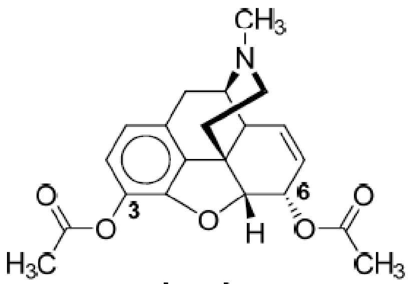
Suggest the structural formulae of the four possible metabolites formed from each compound.

[5]

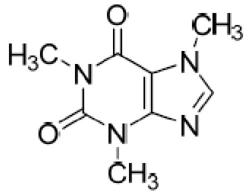
(b)



morphine



heroin



caffeine

When taken orally, heroin undergoes extensive first-pass metabolism to form morphine. When the drug is injected, however, it avoids this first-pass metabolism, entering the brain more rapidly than morphine. Once in the brain, heroin is hydrolysed into 3-monoacetylmorphine (3-MAM), 6-monoacetylmorphine (6-MAM), and morphine which bind to µ-opioid receptors.

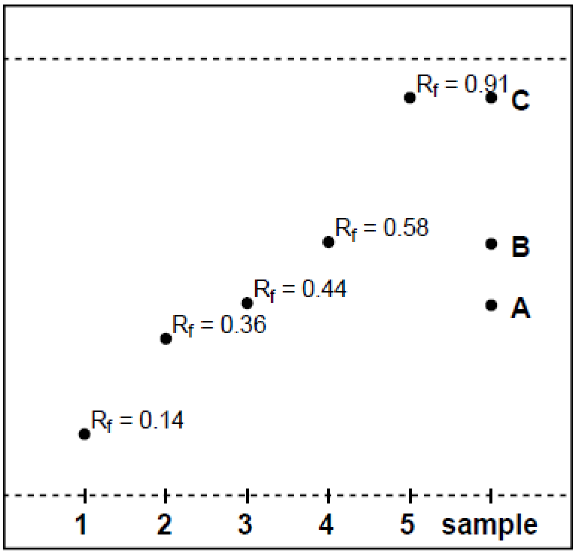
(i) State the pharmacological relationship between heroin and morphine.

(ii) Explain why heroin enters the brain faster than morphine.

(iii) Give the structures of 3-MAM and 6-MAM.

During a police narcotics raid, a sample of impure heroin was seized and analyzed by the Health Science Authority (HSA) chemists. They performed paper chromatography. The chromatogram shown below was developed using a mixture of 2-methylpropan-2-ol and acetic acid in water, buffered at pH 5.1.

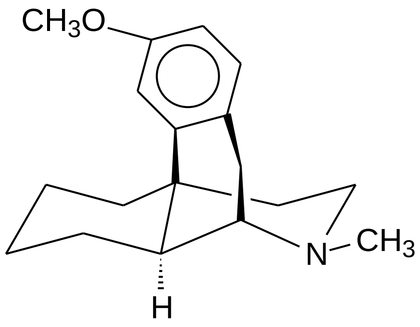
The compounds labeled **1** to **5** are standard samples of caffeine, heroin, morphine, 3-MAM and 6-MAM, not necessarily in the order shown, with their retention factors (R_f) indicated.



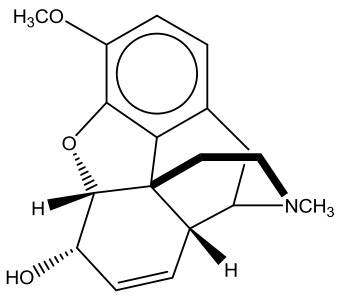
(iv) Identify the three compounds **A**, **B** and **C** found in the heroin sample. Explain how you arrived at your answer.

[7]

(c) Dextromethorphan and codeine are cough suppressants. It affects the signals in the brain that trigger cough reflex. Like morphine, they are narcotic analgesics.



Dextromethorphan



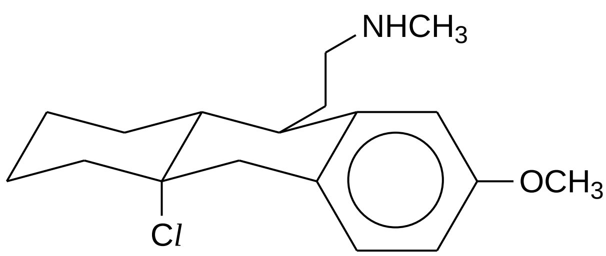
Codeine

(i) Describe briefly how narcotic and non-narcotic analgesics act differently to alleviate pain.

(ii) Compare the solubilities of codeine and dextromethorphan in water.

[4]

(d) Student **A** proposed that heating **precursor A** in hot ethanol will give enantiomerically pure dextromethorphan.



Precursor A

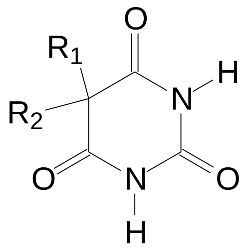
(i) Student **B** disagrees with student **A**. Student **B** argues that since **precursor A** is a tertiary alkyl chloride, it should undergo S_N1 instead of S_N2, and therefore the prepared dextromethorphan should be in racemic form. Do you agree with her? Give reasons for your answer.

(ii) In reality, elimination happens instead of nucleophilic substitution. Elaborate on how isotopic labeling can be used to tell whether the mechanism is E1 or E2.

[4]

[Total: 20]

4 Barbiturates are cyclic imides used as hypnotic and anticonvulsant drugs. They have the following general form:



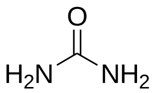
(a) Suggest whether barbiturates are chiral compounds. [2]

(b) A particular type of barbiturate **A** is subjected to NMR analysis.

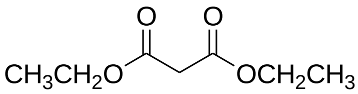
- Under ^{1}H NMR, there is only one peak at about 0.7 ppm.
- Under ^{13}C NMR, there are five peaks at five distinct chemical shifts.

Deduce the exact structure of **A**. [2]

(c) Barbituric acid, $C_4H_4N_2O_3$, a precursor of barbiturates, can be formed from the condensation reaction between urea and diethylmalonate.



urea



diethylmalonate

Draw the structure of barbituric acid. [1]

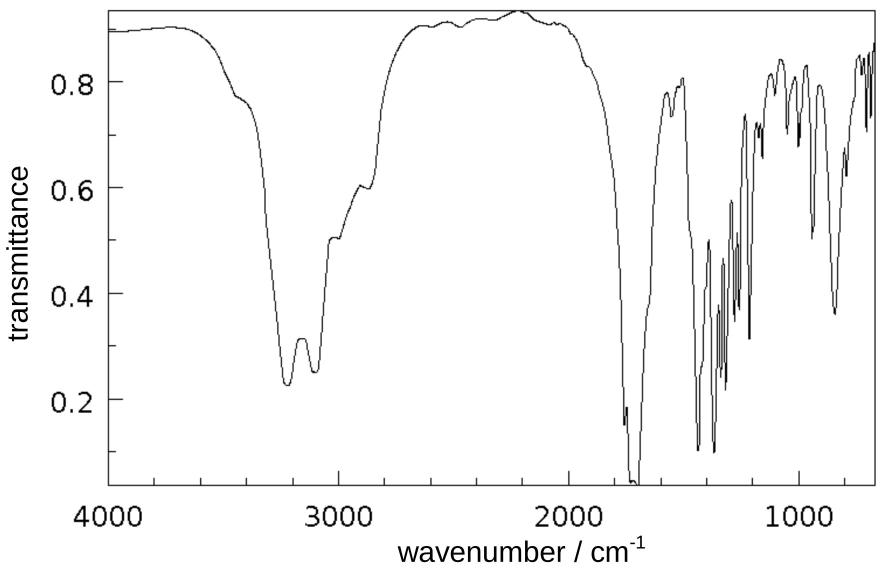
(d) Aprobarbital, a barbituric acid derivative, was used primarily for the treatment of insomnia.

The mass spectrum of aprobarbital shows the following peaks and the molecular ion peak at m/z = 210.

m/z value	**relative abundance**
41	40
167	100
210	2.7
211	0.3

The ^{1}H NMR spectral data and IR spectrum of aprobarbital are given below.

chemical shift / ppm	**integration**	**multiplicity**
1.0	6	doublet
2.3	1	multiplet
2.4	2	doublet
5.0	2	multiplet
5.7	1	multiplet
10.0	2	singlet

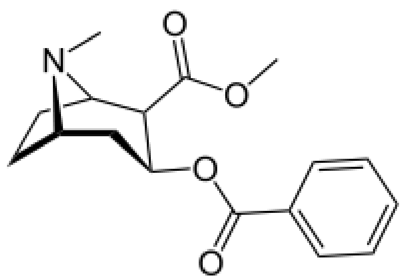


(i) Deduce the number of carbon atoms in aprobarbital.

(ii) Use your answer in **(d)(i)** and the provided spectral information to deduce the structure of aprobarbital, explaining clearly how you arrived at your conclusion.

[6]

(e) Even as early as in the sixteenth century, the Incas chewed the leaves of the coca bush, *Erythroxylon coca*, to combat fatigue. Chemical studies of *Erythroxylon coca* by Friedrich Wöhler in 1862 resulted in the discovery of cocaine as the active component.



cocaine

Nicotine, caffeine and cocaine belong to the class of drugs known as stimulants. Like nicotine, cocaine is found in cigarettes.

(i) Define the term ‘stimulant’.

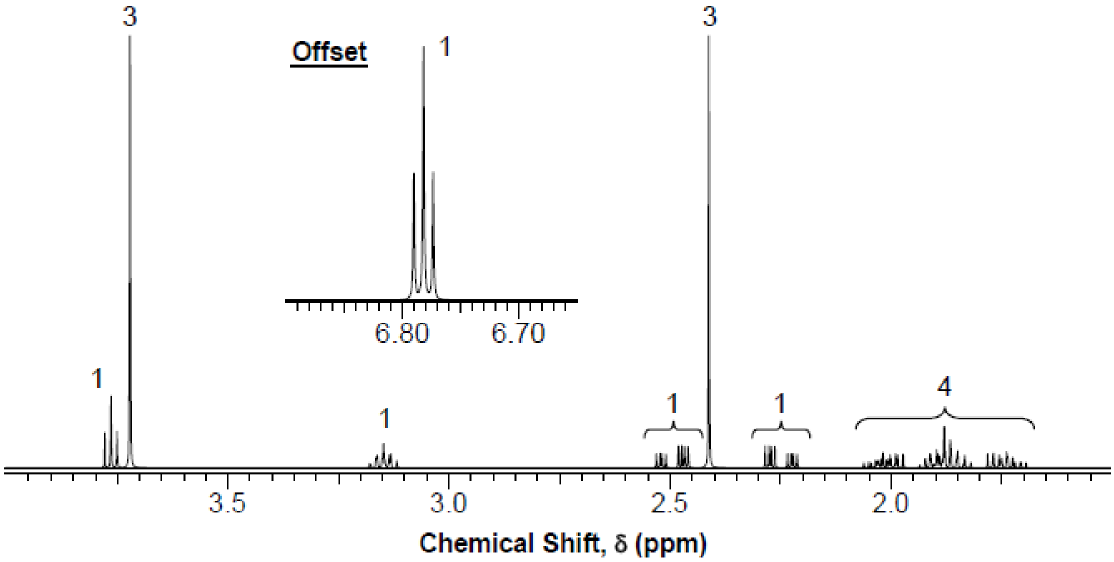
(ii) Suggest a method to make cocaine more water soluble for injection.

(iii) By referring to your answer to **(e)(ii)**, suggest whether drinking a solution of cocaine or smoking cocaine would afford a higher concentration of cocaine in the blood within the same time frame. Explain.

[5]

(f) Methylecgonidine is released into a smoker's body due to the pyrolysis of cocaine. Methylecgonidine is then metabolized into ecgonidine, $C_9H_{13}NO_2$, which is subsequently excreted in urine.

The mass spectrum of methylecgonidine shows a molecular ion peak at m/e 181. The ^{1}H NMR spectrum of methylecgonidine is shown below.

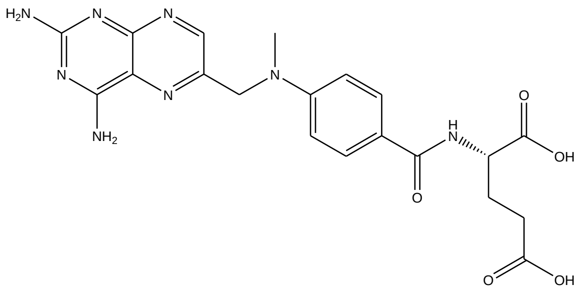


Deduce the structures of ecgonine and methylecgonine.

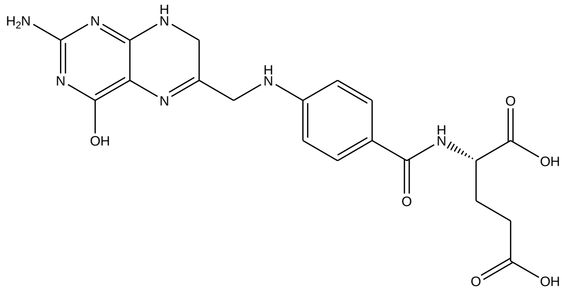
[4]

[Total: 20]

5 Methotrexate is an antibacterial drug designed to interfere with the utility of folic acid in bacteria. After folic acid is synthesized in bacteria, it is stepwise reduced to dihydrofolate. The enzyme dihydrofolate reductase then reduces dihydrofolate to tetrahydrofolate. It is tetrahydrofolate which is the active form of folic acid that possesses biological activity.



Methotrexate



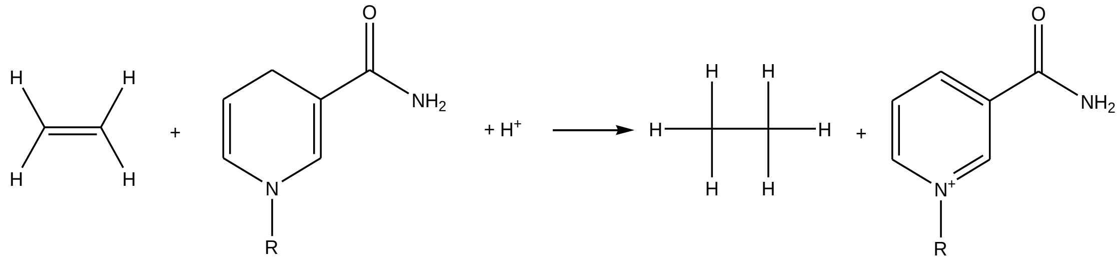
Dihydrofolate

(a) **(i)** Suggest where on the enzyme molecule dihydrofolate binds, and how this occurs at pH 7.4.

(ii) Explain how methotrexate could interfere with the utility of folic acid in bacteria. [4]

(b) Biological systems often employ the coenzyme NADPH as the reducing agent. Upon redox, the target molecule is reduced and NADPH is oxidized to $NADP^+$. Such is the case for dihydrofolate reductase.

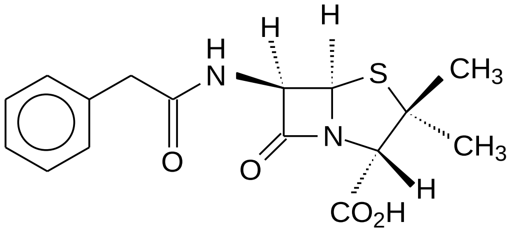
For simplicity, the redox reaction between NADPH and a double bond is shown as:



By drawing in suitable electron-pushing arrows, complete the mechanism for the redox reaction above. [2]

(c) The target of many other antibacterial drugs is the cell wall biosynthetic pathway. Penicillins, a widely prescribed group of antibiotics, act by inhibiting an enzyme involved in the pathway.

The structure of *penicillin G* is shown below.

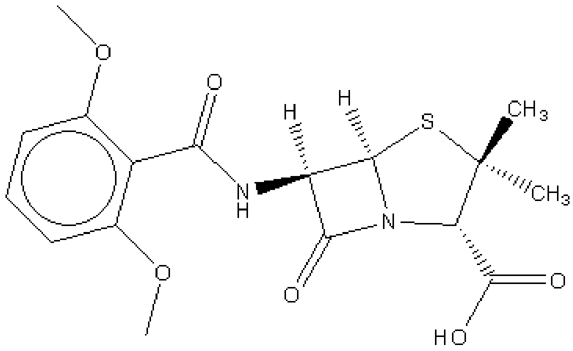


Outline two other ways in which drugs act against bacteria, other than by disrupting folic acid synthesis and by inhibiting cell wall biosynthesis.

[2]

(d) The over-prescription of penicillins have resulted in the evolution of strains of bacteria which are resistant to penicillins such as *penicillin G*.

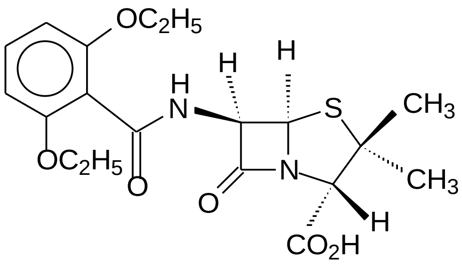
Methicillin was an important penicillin that was synthesized in the 1960s to counter the threat of penicillin-resistant strains of *Staphylococcus aureus*.



(i) Explain why over-prescription of penicillins may have led to resistance.

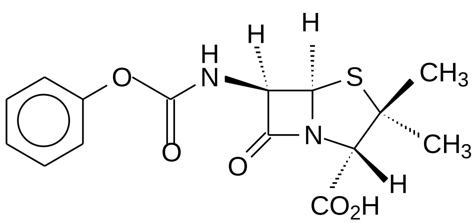
(ii) Explain which structural feature of the *methicillin* molecule allows it to overcome the problem of penicillin resistance.

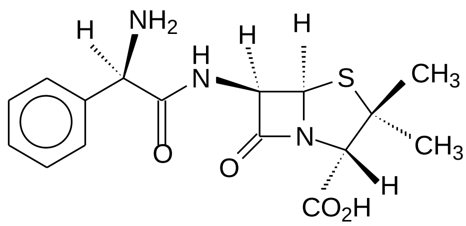
(iii) The following structure is an analogue of *methicillin*.



Suggest, with explanation, how you would expect the antibacterial activity of *methicillin* to compare with this analogue.

(iv) Explain one disadvantage in the use of *penicillin G* and *methicillin* as antibiotics compared to the use of penicillins such as *penicillin V* and *ampicillin*, which respective structures are shown below.





[7]

(e) Compounds **A** and **B** are two fragments formed from *penicillin G* molecule via a reaction route. Analysis of the two fragments was carried out and the following information obtained:

- The identity of compound **A** is found to be

$$HOOC{-}\underset{}{\overset{\displaystyle NH_2\atop |}{CH}}{-}C(CH_3)_2{-}SH$$

- The mass spectrum of compound **B** showed the molecular ion peak at m/e 235, and the M : (M+1) abundance ratio was 22.5 : 3.

- The ^{1}H NMR spectrum of compound **B** showed the following peaks:

chemical shift / ppm	**relative area**	**comment**
1.7	1	singlet
2.0	2	singlet
4.2	1	singlet, disappears on treatment with D_2O
4.5	3	singlet
7.2 – 8.0	5	multiplet
9.1	1	singlet

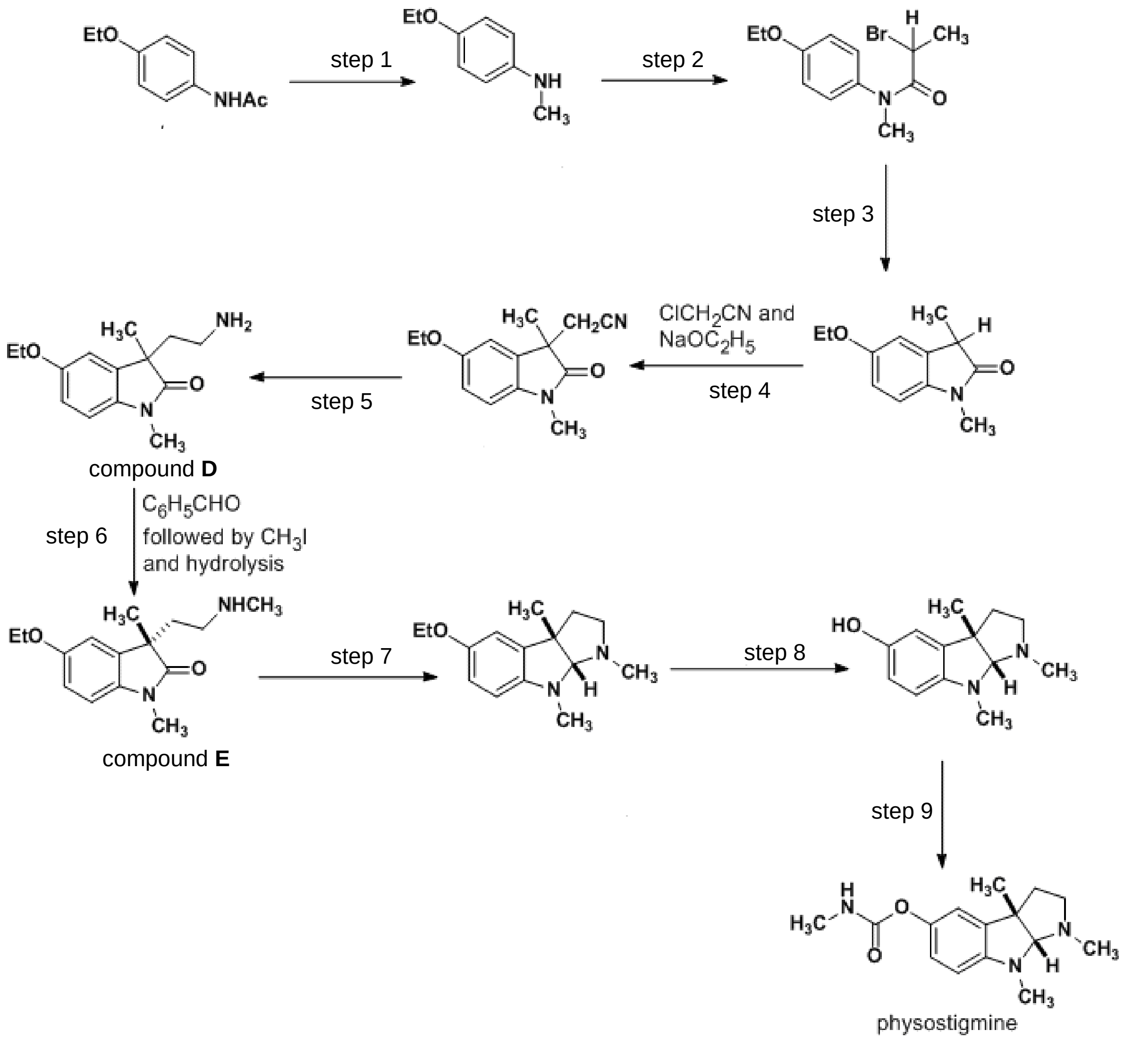
Use these data to deduce a structure for **B**, explaining how you arrived at your conclusions.

[5]

[Total: 20]

6 Physostigmine is commonly used as an antidote for atropine poisoning, where atropine is an acetylcholine antagonist. The drug targets the enzyme acetylcholinesterase which is responsible for hydrolyzing acetylcholine when it has finished transmitting a nerve signal. Recently, it has begun to see more use in Alzheimer's disease and myasthenia gravis where it improves signal transmissions.

The following diagram shows a synthetic route to manufacturing physostigmine. Use the diagram to answer the questions that follow:



(a) Explain what is meant by the term 'antagonist'. [1]

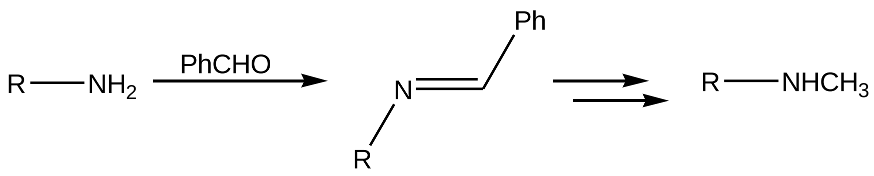
(b) Suggest reagents and conditions for steps 2, 3, 5, 8 and 9. [5]

(c) Outline the mechanism for step 4. [2]

(d) Give two reasons why the removal of the ethyl ether can only be done in step 8. [2]

(e) The reaction of compound **D** with benzaldehyde is shown below.

The R–N=CHPh intermediate reacts with CH_3I and is later hydrolyzed to obtain compound **E**. (Ph represents a phenyl group)

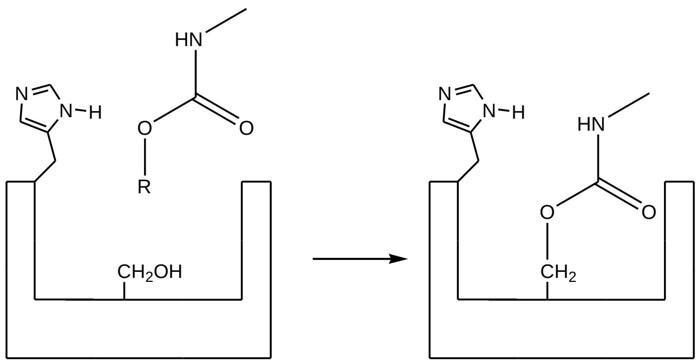


(i) Outline a mechanism for the formation of the R–N=CHPh intermediate.

(ii) Explain why it would not be feasible to simply heat compound **D** with ethanolic CH_3I under reflux to obtain compound **E**.

[3]

(f) Physostigmine reacts with the serine residue (R = $-CH_2OH$) in the active site of acetylcholinesterase according to the diagram below. The reaction is catalyzed by the acidic heterocyclic ring in the vicinity.



(i) Outline the mechanism for the reaction of physostigmine with the serine residue.

(ii) With reference to your answer in **(f)(i)**, explain how physostigmine is an antidote to atropine poisoning.

(iii) The product shown in the diagram above is much more resistant to hydrolysis than its corresponding ester, RCOOR', thus making it useful in small doses. Account for this observation.

[7]

[Total: 20]